Instituto Politécnico de Castelo Branco

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Escola Superior de Tecnologia

Cycle in **Degree** Course
2nd Semester of 3rd Year
Course Unit: **Cloud Computing**
Lecturer: Osvaldo Santos

Practical work on Cosmos DB

Objectives of this sheet

Initial contact with the Cosmos DB database service

Bibliography and useful links

Azure Cosmos DB documentation
https://learn.microsoft.com/en-us/azure/cosmos-db/introduction

Video showing an example of accessing Cosmos DB from Node.JS https://learn.microsoft.com/en-us/shows/docs-azure/use-nodejs-to-connect-and-query-data-from- azure-cosmos-db-sql-api-account

Quickstart for using Cosmos DB in Node.JS
https://learn.microsoft.com/en-us/azure/cosmos-db/nosql/quickstart-nodejs?tabs=azure-portal%2Cconnection-string%2Clinux%2Csign-in-azure-cli

Queries in Cosmos DB for NoSQL https://learn.microsoft.com/en-us/azure/cosmos-db/nosql/query/

Node.JS examples with Cosmos DB
https://learn.microsoft.com/en-us/azure/cosmos-db/nosql/samples-nodejs

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In this assignment you will use your ipcbcampus student account to access Azure

At the end of the job, you must delete all Azure resources so that you can preserve your balance

1) Virtual Network

a) Create a new virtual network, called my-vnet, in the France Central region, resource group class

b) Leave all other vnet features at their default settings

2) Cosmos DB account

a) Create a new Cosmos DB Account, with the "Azure Cosmos DB for NoSQL" API, with the following characteristics:

i) Basics

- Resource Group: class
- Account name: "cosmos-your-name""random number", e.g. cosmos- vitor783457
- Availability zones: disable
- Region: Central France
- Capacity mode: Provisioned throughput
- Apply Free Tier Discount: Apply
- Don't change anything else

ii) Global Distribution

- Don't change anything

iii) Networking

- Connectivity method: Private endpoint
- Allow access from Azure Portal: Allow
- Allow access from my IP: Allow
- Allow Public Network Access: Allow
- Create a private endpoint, with the name "cosmos-endpoint", for the vnet you created in
  1), for the default subnet
- Don't change anything else

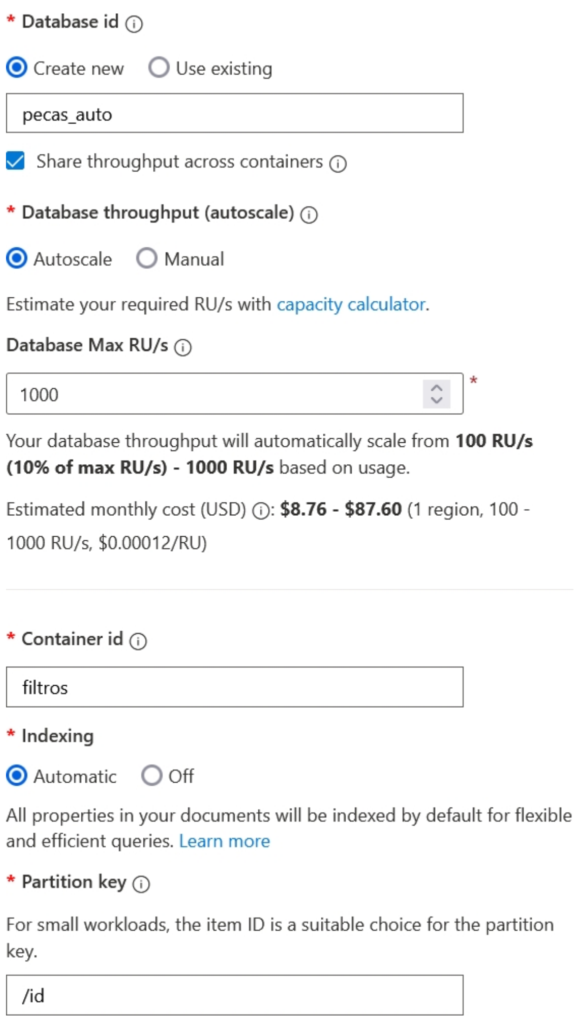
iv) Backup policy
   - Don't change anything

v) Encryption
   - Don't change anything

b) Create the Cosmos DB Account and wait for it to be created. It may take a few minutes.

3) Create database container

a) In the Cosmos DB Account, add a DB container with the following characteristics:



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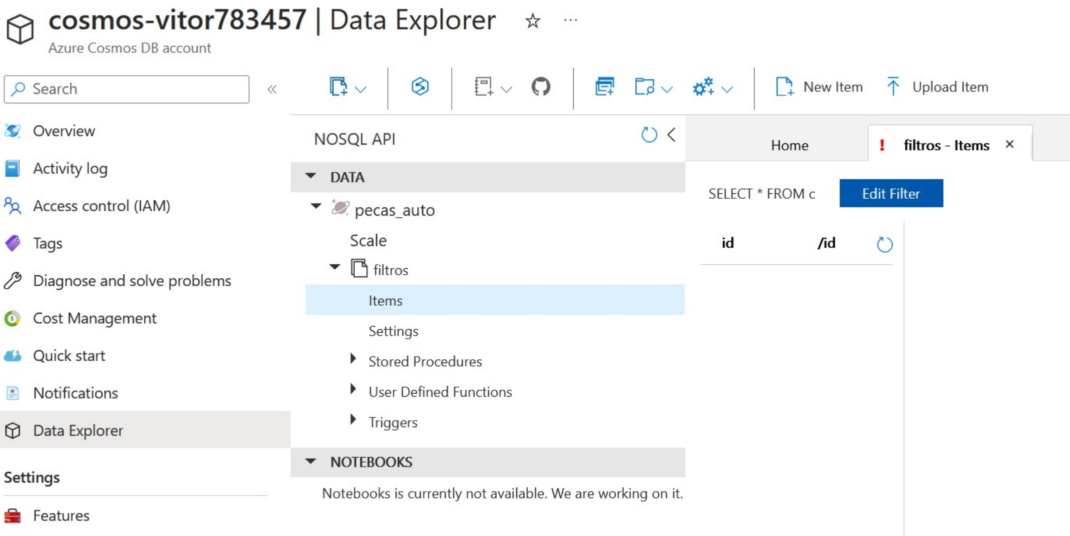
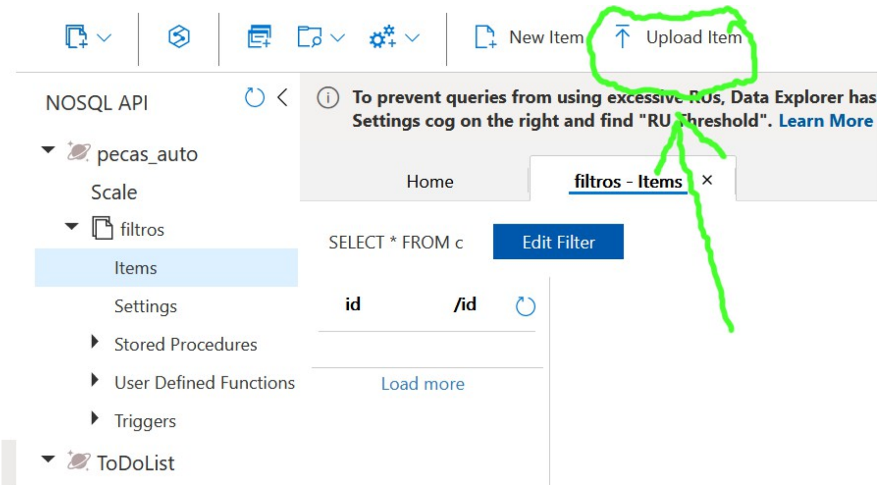
b) Once you've created the container, it should look like this:



c) Now upload the items to the container. The items are provided by the lecturer in a json file, which is provided in moodle. Tip:



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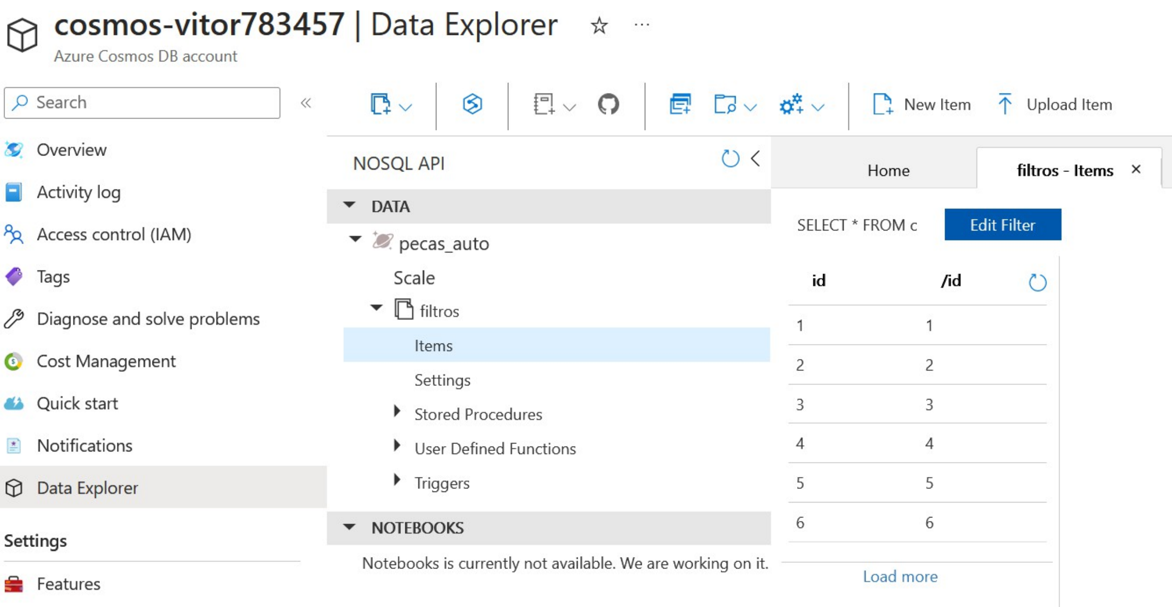
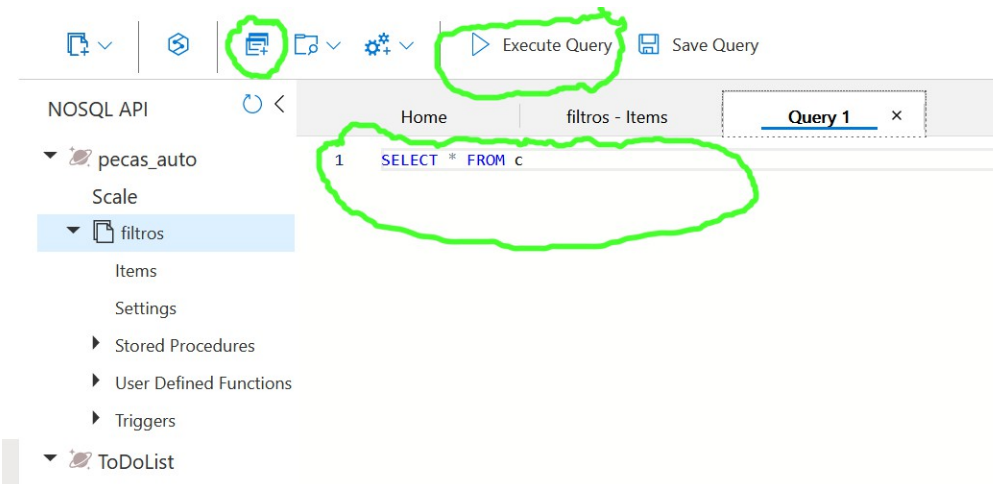
d) After uploading, you should have a configuration similar to this one, with 6 items:



e) Try looking at the content of each item by clicking on the corresponding line.

f) You can test queries directly on the portal, in the area shown in the following figure:




```
Tip on the format of queries:
```

https://learn.microsoft.com/en-us/azure/cosmos-db/nosql/query/

g) Now try some local queries to get:

   i) The list of filters from the manufacturer "Lucas"

   ii) The list of filters for Renault cars

   iii) EAN code and price of filters from the manufacturer "Delphi" for vehicles of the brand "Renault"

   iv) The EAN code, the manufacturer's name and the price of filters for "Renault" brand vehicles costing less than €20.

   v) The EAN code, the name of the filter manufacturer and the quantity of products whose stock is below the minimum stock.

4) Access to the database from a virtual machine

Note: this question has a high degree of difficulty. If you can't solve it, the teacher will provide the code with the solution.

a) Create a VM in the region **<u>and vnet you created in 1)</u>**, with Ubuntu Server 22, with automatic installation of Node.js version 20. The cloud-init file for this VM is provided by the teacher.

b) In this VM, create a node.JS script called filters.js , which shows the list of filters in the database (only EAN code, manufacturer, stock_existing and stock_minimum).

```
Tip: See the links on the first page
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c) Test the script in the VM console

5) Serverless Cosmos DB Trigger function

Note: this question has a high degree of difficulty. If you can't solve it, the teacher will provide the code with the solution.

a) Create a serverless function in Node.JS that is triggered from the Cosmos DB database (whenever there is a change in an item).

b) The function should check whether the existing stock is less than the minimum stock. If so, it should order the number of parts from the manufacturer so that the stock is double the minimum_stock. The order should be a message sent to a message queue.